Research Areas and Simulation in Cloud Computing

This article looks at the backbone technologies in cloud computing, and suggests some areas of research in this field as well as a few resources.

Cloud computing is one of the emergent domains in which remote resources are used on the basis of demand, even without the physical infrastructure at the client end. In cloud computing, the actual resources are installed and deployed at remote locations. This article focuses on the guidelines for research scholars and practitioners in the domain of cloud computing and related technologies.

Backbone technologies in cloud computing

Virtualisation

Virtualisation is the major technology that works with cloud computing. An actual cloud is implemented with the use of virtualisation technology. In cloud computing, dynamic virtual machines are created to provide access to actual infrastructure to an end user or developer at another remote location.

A virtual machine or VM is the software implementation of any computing device, machine or computer that executes the series of instructions or programs as a physical (actual) machine. When a user or developer works on a VM, the resources, including all programs installed on the remote machine, are accessible using a specific set of protocols. Here, for the end user of the cloud service, the VM acts like the actual machine.

Prominent virtualisation software

Prominent virtualisation software on Windows and Linux host OSs are:

- *Windows as the host OS*
  - VMware Workstation (any guest OS)
  - Virtual Box (any guest OS)
  - Hyper-V (any guest OS)
- *Linux as the host OS*
  - VMware Workstation
  - Microsoft Virtual PC
  - VMLite Workstation
  - VirtualBox
  - Xen

A hypervisor or virtual machine monitor (VMM) is a piece of computer software, firmware or hardware that creates and runs virtual machines. A computer on which a hypervisor is running one or more virtual machines is defined as a host machine. Each virtual machine is called a guest machine. The hypervisor presents the guest OSs with a virtual operating platform and manages their execution. Multiple instances of a variety of OSs may share virtualised hardware resources.

Hypervisors of Type 1 (bare metal installation) and Type 2 (hosted installation)

Type 1 hypervisors are used in the implementation and deployment of cloud services, and they are associated with the